Sheaves and Real-World Information

An Introduction and Reading Guide

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SUMMARY. Sheaves were introduced in the 1940s to pass from local to global information in topology, and for sixty years they belonged almost entirely to pure mathematics. Since about 2010 a finite, computable version of the theory has been used to integrate measurements from real sensors, to describe entropy and the consistency of probabilistic data, and to build neural networks. These notes recall what a sheaf is and where the idea came from, explain what changed to make sheaves usable on actual data, state the main definitions and results without proofs, and end with an ordered reading guide to the literature.

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1. The idea in one paragraph

A sheaf assigns data to the parts of a space and records how data on a larger part restrict to a smaller one, in such a way that compatible local data determine a unique global datum. Its cohomology measures the failure of that last step: the local data that cannot be assembled, and the ambiguity when they can. For a mathematician the standard examples are functions, differential forms, and solutions of differential equations. The applied theory reads the same structure differently. The "space" is a network of sensors, variables, or agents; the data on each part are measurements, probability

distributions, or feature vectors; the restriction maps say how one observation is to be compared with another; a global section is a globally consistent account of all the observations; and cohomology, or a metric refinement of it, quantifies inconsistency. Everything below elaborates this dictionary.

2. What a sheaf is

2.1. Presheaves and sheaves

Let X be a topological space and $\mathrm{Op}(X)$ its poset of open sets, ordered by inclusion.

Definition 2.1 (Presheaf). A *presheaf* of sets on X is a functor $\mathcal{F} : \mathrm{Op}(X)^{\mathrm{op}} \to \mathbf{Set}$. It assigns a set $\mathcal{F}(U)$ of *sections* to each open U, and to each inclusion $V \subseteq U$ a *restriction map* $\rho^U_V : \mathcal{F}(U) \to \mathcal{F}(V)$, written $s \mapsto s|_V$, such that $\rho^U_U = \mathrm{id}$ and $\rho^V_W \circ \rho^U_V = \rho^U_W$ for $W \subseteq V \subseteq U$. Replacing **Set** by abelian groups, vector spaces, or rings gives presheaves with values in those categories.

Definition 2.2 (Sheaf). A presheaf $\mathcal{F}$ is a *sheaf* if for every open U and every open cover $U = \bigcup_i U_i$:

(i) *(locality)* if $s, t \in \mathcal{F}(U)$ satisfy $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$;

(ii) *(gluing)* if $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then there is $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all i.

Equivalently, the diagram $\mathcal{F}(U) \to \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$ is an equalizer.

Example 2.3. Continuous real functions, $U \mapsto C(U, \mathbb{R})$, form a sheaf, as do smooth functions, holomorphic functions, sections of a vector bundle, and solutions of a linear differential equation. Bounded functions form a presheaf that is not a sheaf: bounded functions on the members of a cover glue to a function that need not be bounded. Constant functions likewise fail to glue on a disconnected set, whereas *locally* constant functions form the *constant sheaf* $\underline{A}$.

Definition 2.4 (Stalks). The *stalk* of $\mathcal{F}$ at $x \in X$ is $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$; its elements are *germs* of sections at x. A morphism of sheaves is an isomorphism if and only if it induces isomorphisms on all stalks.

Proposition 2.5 (Sheafification). *The inclusion of sheaves into presheaves has a left adjoint $\mathcal{F} \mapsto \mathcal{F}^+$, which preserves stalks. Concretely, $\mathcal{F}^+(U)$ consists of the families of germs $(s_x)_{x \in U}$ that are locally induced by sections of $\mathcal{F}$.*

2.2. Cohomology

For a sheaf of abelian groups the global-sections functor $\Gamma(X, -) : \mathcal{F} \mapsto \mathcal{F}(X)$ is left exact but not exact: a surjection of sheaves need not be surjective on global sections.

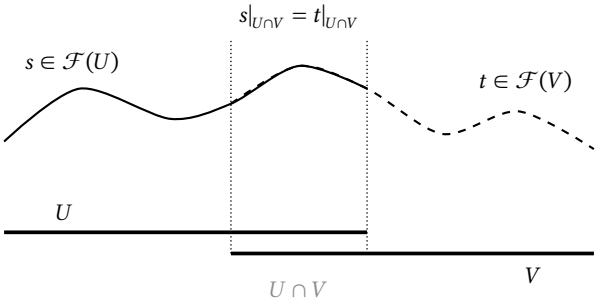


FIGURE 1. Gluing. Two sections that agree on the overlap determine exactly one section over $U \cup V$. Locality gives uniqueness, gluing gives existence.

DEFINITION 2.6 (Sheaf cohomology). $H^i(X, \mathcal{F})$ is the i-th right derived functor of $\Gamma(X, -)$ applied to $\mathcal{F}$. Thus $H^0(X, \mathcal{F}) = \mathcal{F}(X)$, and a short exact sequence of sheaves induces a long exact sequence in cohomology.

The working interpretation is that H^0 is the space of global solutions and H^1 classifies the obstructions to assembling local solutions into a global one. In practice cohomology is computed from a cover. For an open cover $\mathcal{U} = \{U_i\}$, the Čech complex has $\check{C}^k(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \cdots < i_k} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_k})$ with the alternating-sum differential.

THEOREM 2.7 (Leray's theorem on acyclic covers). *If $H^q(U_{i_0} \cap \cdots \cap U_{i_k}, \mathcal{F}) = 0$ for all $q > 0$ and all finite intersections of members of $\mathcal{U}$, then $\check{H}^k(\mathcal{U}, \mathcal{F}) \cong H^k(X, \mathcal{F})$ for all k.*

Theorem 2.7 is the reason the theory can be made finite: with a good cover, cohomology is the cohomology of an explicit complex indexed by a combinatorial object, the nerve of the cover. Section 5 takes that combinatorial object as the starting point.

3. Where sheaves came from

Sheaves were not created by Grothendieck, although the association is understandable: he rebuilt the theory in the form now taught. The history has three steps (Miller 2000; Gray 1979; Houzel 1990).

Leray. Jean Leray, an analyst known for his work on the Navier–Stokes equations, spent 1940 to 1945 as a prisoner of war in the officers' camp Oflag XVII-A in Austria. To avoid work that could serve the war effort he turned to algebraic topology, and there he devised three things at once: sheaves (*faisceaux*), their cohomology, and spectral sequences. His aim was topological and quite specific: to relate the cohomology of a space to that of the fibres of a map defined on it, with fixed-point theorems for equations in analysis as the motivation. The work became known only after the war, and several more years passed before it was widely understood.

Cartan, Oka, Serre. In his Paris seminar of 1948–51 Henri Cartan reformulated Leray's theory in terms of open sets, with the étalé-space definition due to Michel

Lazard, and it became the natural language for several complex variables: Oka's coherence theorem and Cartan's Theorems A and B are statements about sheaf cohomology. In 1955 Serre's *Faisceaux algébriques cohérents* carried the method into algebraic geometry, where there are no small open sets and sheaf cohomology had to do the work that classical topology could not (Serre 1955).

Grothendieck. The 1957 Tôhoku paper placed sheaf cohomology within homological algebra: abelian categories, enough injectives, and derived functors, so that cohomology is defined on any space for any sheaf (Grothendieck 1957). Schemes followed, and then, in the early 1960s, sites and toposes, which free the notion of sheaf from topological spaces altogether. Godement's book fixed the textbook form of the classical theory (Godement 1958), and the microlocal theory of Sato and Kashiwara extended it to analysis (Kashiwara and Schapira 1990).

4. From pure mathematics to data

It is not accurate to say that sheaves lay unused. From 1950 onwards they were central to algebraic geometry, complex analysis, and the theory of differential equations. What arrived late was their use as a model for *data*, and there were two reasons for the delay. The first is computational: a sheaf on a topological space involves infinitely many open sets and typically infinite-dimensional spaces of sections, so that nothing can be computed from measurements. The second is that the people who had data, and the people who knew sheaves, were not the same people.

Three developments, at first unconnected, prepared the applied theory.

- In probability, Vorob'ev (1962) characterized when a consistent family of marginal distributions extends to a joint distribution. This is a gluing theorem stated without sheaves (Theorem 6.3 below).
- In computer science, Goguen proposed from the 1970s that systems of interacting objects be modelled by sheaves, with behaviour over time as sections (Goguen 1992).
- In topology, Shepard (1985) showed that sheaves constructible with respect to a cell structure are equivalent to a finite, purely combinatorial object: a *cellular sheaf*.

The last is the decisive technical step. A cellular sheaf is a finite diagram of vector spaces and linear maps, and its cohomology is the cohomology of a finite complex of matrices. Around 2010 this was taken up in applied topology, principally by Robert Ghrist and his students, and within a few years the three lines of Section 6 appeared: sensor integration (Robinson 2017), contextuality and entropy (Abramsky and Brandenburger 2011; Baudot and Bennequin 2015), and, from 2020, machine learning (Hansen and Gebhart 2020; Bodnar et al. 2022). Table 1 gives the chronology.

TABLE 1. A timeline. The upper part is the classical theory; the lower part, its use for data.

1940–45	Leray, as a prisoner of war, devises sheaves, sheaf cohomology, and spectral sequences.
1948–51	Cartan's seminar: open sets, étalé spaces (Lazard), application to several complex variables.
1955	Serre, *Faisceaux algébriques cohérents*: sheaves enter algebraic geometry.
1957	Grothendieck, the Tôhoku paper: derived functors; then schemes, sites, and toposes.
1958	Godement's textbook.
1990	Kashiwara and Schapira, *Sheaves on Manifolds*: the microlocal theory.
1962	Vorob'ev: when consistent marginals extend to a joint distribution.
1985	Shepard's thesis: cellular sheaves.
1992	Goguen: sheaf semantics for concurrent interacting objects.
2011	Ghrist and Hiraoka: network coding as sheaf cohomology. Abramsky and Brandenburger: contextuality as the absence of a global section.
2014	Curry's thesis systematizes cellular sheaves and cosheaves. Robinson, *Topological Signal Processing*. Ghrist, *Elementary Applied Topology*.
2015	Baudot and Bennequin: Shannon entropy as a cohomology class.
2017	Robinson: sheaves as the canonical data structure for sensor integration.
2019	Hansen and Ghrist: the sheaf Laplacian and its spectral theory.
2020	Robinson: the consistency radius. Joslyn et al.: sheaf-based tracking on field data. Vigneaux: information cohomology in general. Hansen and Gebhart: sheaf neural networks.
2022	Bodnar et al.: neural sheaf diffusion, the first results on standard benchmarks.
2025	Ayzenberg et al.: a 117-page survey of sheaves in deep learning.

5. Cellular sheaves: the computable version

Let X be a regular cell complex (a graph or a simplicial complex will do) and write $\sigma \trianglelefteq \tau$ when the cell σ is a face of τ.

Definition 5.1 (Cellular sheaf). A *cellular sheaf* $\mathcal{F}$ of vector spaces on X assigns a vector space $\mathcal{F}(\sigma)$, the *stalk*, to each cell σ, and a linear *restriction map* $\mathcal{F}_{\sigma \trianglelefteq \tau} : \mathcal{F}(\sigma) \to \mathcal{F}(\tau)$ to each incidence $\sigma \trianglelefteq \tau$, such that $\mathcal{F}_{\sigma \trianglelefteq \sigma} = \mathrm{id}$ and $\mathcal{F}_{\tau \trianglelefteq v} \circ \mathcal{F}_{\sigma \trianglelefteq \tau} = \mathcal{F}_{\sigma \trianglelefteq v}$. In other words, $\mathcal{F}$ is a functor from the face poset of X to vector spaces.

The maps run from faces to cofaces, which is the direction of restriction once cells are identified with their open stars: the star of an edge is contained in the star of each of its endpoints. A cellular sheaf is equivalent to a sheaf in the sense of Definition 2.2 on the face poset with its Alexandrov topology (Curry 2014).

Definition 5.2 (Sections and cohomology). A *global section* of $\mathcal{F}$ is a choice $x_\sigma \in \mathcal{F}(\sigma)$ for every cell such that $\mathcal{F}_{\sigma \trianglelefteq \tau} x_\sigma = x_\tau$ whenever $\sigma \trianglelefteq \tau$. Let $C^k(X;\mathcal{F}) = \bigoplus_{\dim \sigma = k} \mathcal{F}(\sigma)$ and define $\delta : C^k \to C^{k+1}$ by $(\delta x)_\tau = \sum_{\sigma \trianglelefteq \tau} [\sigma : \tau]\, \mathcal{F}_{\sigma \trianglelefteq \tau} x_\sigma$, where $[\sigma : \tau] = \pm 1$ is the incidence number. Then $\delta^2 = 0$, and $H^k(X;\mathcal{F}) = \ker \delta^k / \operatorname{im} \delta^{k-1}$.

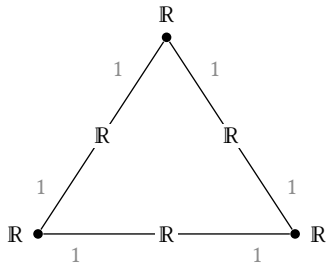
Proposition 5.3. *$H^0(X;\mathcal{F}) = \ker \delta^0$ is the space of global sections. On a graph, with an edge e oriented from u to v, $(\delta x)_e = \mathcal{F}_{v \trianglelefteq e} x_v - \mathcal{F}_{u \trianglelefteq e} x_u$, so a 0-cochain is a section exactly when the two endpoints of every edge agree after restriction to the edge.*

Figure 2 shows the smallest example in which local consistency does not yield a global section. When the stalks carry inner products, the adjoint of δ is available.

Definition 5.4 (Sheaf Laplacian). The *sheaf Laplacian* of $\mathcal{F}$ is $L_{\mathcal{F}} = \delta^* \delta : C^0 \to C^0$, and more generally $\Delta^k = (\delta^k)^* \delta^k + \delta^{k-1} (\delta^{k-1})^*$. For the constant sheaf $\underline{\mathbb{R}}$ on a graph, $L_{\mathcal{F}}$ is the usual graph Laplacian.

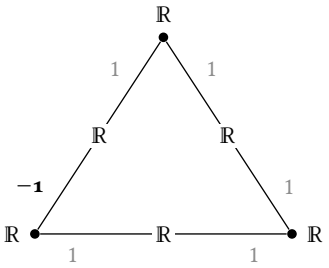
Theorem 5.5 (Hansen and Ghrist 2019). *For a cellular sheaf of finite-dimensional inner-product spaces, $\ker \Delta^k \cong H^k(X;\mathcal{F})$. In particular $\ker L_{\mathcal{F}}$ is the space of global sections, and the solution of the heat equation $\dot{x} = -L_{\mathcal{F}}\, x$ converges, as $t \to \infty$, to the orthogonal projection of $x(0)$ onto the space of global sections.*

Theorem 5.5 turns the search for the consistent account nearest to given data into a linear dynamical system that each node can run using only its neighbours. Both the fusion algorithms of Section 6.1 and the neural networks of Section 6.3 rest on it.



(a) constant sheaf

$H^0 = \mathbb{R},\ H^1 = \mathbb{R}$



(b) one map reversed

$H^0 = 0,\ H^1 = 0$

FIGURE 2. Two cellular sheaves on a triangle, with stalk $\mathbb{R}$ on every vertex and edge. The number at each end of an edge is the restriction map from that vertex to the edge. (a) All maps are the identity: sections are the constant assignments. (b) Reversing the sign of a single map leaves every local condition satisfiable but removes all non-zero global sections: going once round the cycle returns the negative of the starting value.

6. THREE USES

6.1. Integrating measurements

Robinson (2017) argues that any reasonable set of axioms for combining heterogeneous sources leads to a sheaf. The base is a finite topological space, or a cell complex, whose points are the sensors and their overlaps; the stalk over a sensor is the set in which its readings take values; the stalk over an overlap is a common set in which the readings of the overlapping sensors can be compared; and the restriction maps perform the conversion. The actual readings form an *assignment*: one element of each stalk, with no consistency required.

DEFINITION 6.1 (Consistency radius, Robinson 2020). Let $\mathcal{F}$ be a sheaf of pseudometric spaces on a finite space and a an assignment, with $a(U) \in \mathcal{F}(U)$ for each open U. Its *consistency radius* is

$$c(a) = \sup_{V \subseteq U} d_V\big(\rho_V^U\, a(U),\, a(V)\big).$$

PROPOSITION 6.2. *When the stalks are metric spaces, $c(a) = 0$ if and only if a is a global section. The consistency radius is continuous in the assignment, and thresholding it yields a filtration of the base space that is stable under perturbation of the data.*

Data fusion then becomes an optimization problem: find the global section nearest to the assignment, with the consistency radius as a certified bound on the residual disagreement. Two treatments with real measurements are worth reading for the modelling as much as for the mathematics: the location of a hidden radio transmitter from bearings and signal strengths (Robinson 2019), and the tracking of wild black bears from several kinds of sensor at once (Joslyn et al. 2020).

6.2. Information theory

The marginal problem. Let variables be the vertices of a simplicial complex K whose faces are the sets of variables observed jointly. Assigning to each face the probability distributions on its variables, with marginalization as restriction, gives a presheaf. A family of distributions, one per maximal face, is *consistent* if its members agree on common marginals.

Theorem 6.3 (Vorob'ev 1962). *Every consistent family of distributions on the faces of K is the family of marginals of a single joint distribution if and only if K is regular, that is, acyclic in the sense that it can be reduced to a point by successively removing faces attached along a single face.*

When K is not regular there are consistent families with no joint distribution: local sections with no global section. Abramsky and Brandenburger (2011) showed that this is exactly the structure of non-locality and contextuality in quantum mechanics, and Abramsky (2013) that the same question arises in database theory as the existence of a universal relation for a set of tables. Čech cohomology supplies computable witnesses of the obstruction (Abramsky, Barbosa, et al. 2015).

Entropy. Baudot and Bennequin (2015) define a cohomology theory for an *information structure*, a category of observables ordered by refinement, with coefficients in functions of probability distributions on which observables act by conditional expectation.

Theorem 6.4 (Baudot and Bennequin 2015; Vigneaux 2020). *The 1-cocycle condition of information cohomology is the chain rule $f(XY) = f(X) + X.f(Y)$. Under mild non-degeneracy assumptions on the information structure, the first cohomology group is one-dimensional and is generated by Shannon entropy. For the action deformed by a parameter $\alpha \neq 1$, it is generated by the Tsallis α-entropy.*

Entropy is thus characterized by a topological invariant instead of by a list of axioms. Two related results complete the picture: information flows in a network code are the degree-zero cohomology of a sheaf on the network (Ghrist and Hiraoka 2011), and belief propagation in graphical models is a diffusion whose fixed points are described homologically (Peltre 2020).

6.3. Machine learning

A graph neural network diffuses node features with the graph Laplacian, which by the remark above is the Laplacian of the constant sheaf. Its limit is the constant section, which is the phenomenon known as oversmoothing, and it implicitly assumes that neighbours should agree, which fails on *heterophilic* graphs. Hansen and Gebhart (2020) proposed replacing the constant sheaf by a general cellular sheaf, and Bodnar et al. (2022) made the restriction maps learnable. Schematically, a layer is

$$X_{t+1} = X_t - \sigma\big(L_{\mathcal{F}(t)}\,(I \otimes W_1)\,X_t\,W_2\big),$$

where the sheaf $\mathcal{F}(t)$ is produced from the features by a small network. By Theorem 5.5 the diffusion converges to the global sections of the learned sheaf, which need not be constant. Bodnar et al. prove that suitable non-trivial sheaves separate classes of nodes, in the diffusion limit, in settings where the constant sheaf cannot, and that richer families of restriction maps separate more classes.

Three further references mark the range of the subject. Gebhart et al. (2023) show that standard knowledge-graph embeddings are sheaves over the graph's schema, so that a consistent embedding is a global section. Belfiore and Bennequin (2021) place every feed-forward architecture in a Grothendieck topos, with invariances as stacks; this is the most abstract and least tested proposal. Ayzenberg et al. (2025) survey the whole area.

7. Reading guide

The order below assumes a reader who knows algebraic topology and homological algebra but has not worked with sheaves on data.

7.1. A first pass

1. Robinson (2019). A short expository article built around one worked problem with real measurements. It shows the dictionary of Section 1 in action.
2. Curry (2014), the introduction and the chapters on cellular sheaves. This is the place to learn Definition 5.1, its equivalence with constructible sheaves, and the dual notion of a cosheaf.
3. Hansen and Ghrist (2019), for the Laplacian and Theorem 5.5. It is the technical basis of everything computational.

7.2. Three tracks

Data fusion
Robinson (2017) for the axiomatic argument; Robinson (2020) for the consistency radius and its stability; Joslyn et al. (2020) for a complete application; Robinson (2014) for the book-length account, including sampling and filtering.

Information theory
Abramsky and Brandenburger (2011), then Vorob'ev (1962) for the theorem behind it; Baudot and Bennequin (2015) for the idea and Vigneaux (2020) for the rigorous general framework; Peltre (2020) for message passing.

Machine learning
Bodnar et al. (2022), which is self-contained; Gebhart et al. (2023) for knowledge graphs; Ayzenberg et al. (2025) as the reference survey. Hansen and Ghrist (2021) applies the same diffusion to opinion dynamics and is a good source of intuition.

7.3. Background and history

For the classical theory, Bredon (1997) or the first chapters of Kashiwara and Schapira (1990); for the categorical view, Mac Lane and Moerdijk (1992). Rosiak (2022) is an introduction written around examples from outside pure mathematics, and Ghrist (2014) places sheaves within applied topology as a whole. For the history, Miller (2000) on Leray and Gray (1979) for the period up to the 1970s.

7.4. Software

Two open libraries implement the constructions above: `pysheaf` (Python, by Robinson; sheaves on finite spaces, consistency radius, and fusion by optimization) and `CellularSheaves.jl` (Julia, in the AlgebraicJulia project; cellular sheaves and sheaf Laplacians).

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